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OSCILLATION, FREQUENCY GENERATION, AND AC SIGNAL GENERATION IN LINEAR ELECTRIC CIRCUITS

*A Research Report on the Theory and Application of Oscillatory and Periodic Signal
Phenomena in Circuit Analysis*

Course: Electrical Engineering & Circuit Analysis (EE & CA)

Course Code: CSE 1205

Department of Computer Science and Engineering

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Primary References

Alexander, C. K., & Sadiku, M. N. O. — Fundamentals of Electric Circuits
Boylestad, R. L. — Introductory Circuit Analysis

Abstract

Oscillation, frequency generation, and AC signal generation are three closely related phenomena that together form the foundation of communication systems, instrumentation, power electronics, and timing circuits. This report examines the subject from the standpoint of fundamental circuit theory rather than as isolated electronics tricks. It begins with the natural, source-free response of second-order RLC networks to explain why and how oscillation arises physically from the exchange of energy between an inductor and a capacitor. It then extends this idea to resonance and to the design of frequency-selective and frequency-generating networks, including the feedback condition (the Barkhausen criterion) required for an electronic circuit to sustain continuous oscillation. Finally, it connects these ideas to the generation, representation, and practical production of alternating-current (AC) signals using phasors and sinusoidal steady-state analysis. Throughout, the discussion is grounded in, and cross-referenced to, *Fundamentals of Electric Circuits* by Charles K. Alexander and Matthew N. O. Sadiku and *Introductory Circuit Analysis* by Robert L. Boylestad, with chapter and page references supplied for further reading and verification.

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1. Introduction

Every electronic system that communicates, computes, or measures something time-varying depends on a stable, repeatable alternating signal. A mobile phone needs a precise radio-frequency carrier; a microprocessor needs a stable clock; a power grid needs a 50/60 Hz sinusoid; a laboratory needs a function generator capable of producing sine, square, or triangular waves at a chosen frequency. All of these depend on the same underlying circuit principle: a network containing energy-storage elements (inductors and capacitors) can be made to exchange energy back and forth at a characteristic rate, and this natural rate of exchange can be shaped, sustained, and exploited to generate a usable AC signal.

This report is organized around three interlinked ideas that are often taught separately but are, in fact, a single continuous story. Section 2 explains oscillation as the natural (source-free) response of second-order circuits — the underlying physics of why a circuit “wants” to oscillate. Section 3 explains frequency generation — how resonance fixes the rate of that oscillation, and how active feedback circuits keep the oscillation from dying out due to resistive losses. Section 4 explains AC signal generation — how the resulting periodic waveform is represented, analyzed, and produced for practical use. Section 5 draws the three ideas together, Section 6 surveys real engineering applications, and Section 7 concludes the report. Page-level references to both prescribed textbooks are provided throughout and consolidated in the reference list at the end of the report.

2. Oscillation in Electric Circuits

2.1 The Physical Basis of Oscillation

Oscillation in an electric circuit has the same physical root as oscillation in a mechanical pendulum or a mass on a spring: a repeated exchange of energy between two different storage forms. In a circuit, an inductor stores energy in its magnetic field ($w_L = \frac{1}{2} L i^2$) while a capacitor stores energy in its electric field ($w_C = \frac{1}{2} C v^2$). If an inductor and capacitor are connected together with no resistance present, energy delivered to the capacitor is later transferred to the inductor and back again indefinitely, producing a sinusoidal current and voltage that never decays. Any real circuit also contains resistance, which dissipates energy as heat and removes it from this exchange, so a purely passive RLC network can only oscillate temporarily before the oscillation dies away. This is precisely why a sustained oscillator needs an external active element to replace the energy lost to resistance every cycle, a point developed further in Section 3.

2.2 The Source-Free Series RLC Circuit

Alexander and Sadiku develop this idea formally through the source-free series RLC circuit (Fundamentals of Electric Circuits, Chapter 8, “Second-Order Circuits,” Section 8.3, p. 317). Applying Kirchhoff’s voltage law to a series loop of R, L, and C produces a second-order homogeneous differential equation in the loop current. Solving it leads to a characteristic equation whose roots are

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}, \quad \alpha = R / 2L, \quad \omega_0 = 1 / \sqrt{LC}$$

where α is the neper (damping) frequency and ω_0 is the undamped natural (resonant) angular frequency of the circuit. Depending on the relative size of α and ω_0 , the response falls into one of three regimes:

- Overdamped ($\alpha > \omega_0$): the response decays to zero without oscillating, through two real, distinct roots.
- Critically damped ($\alpha = \omega_0$): the fastest possible non-oscillatory decay, with two equal real roots.
- Underdamped ($\alpha < \omega_0$): the roots become complex, $s_{1,2} = -\alpha \pm j\omega_d$, and the response is a decaying sinusoid — a genuine, though damped, oscillation.

The damped oscillation frequency is given by $\omega_d = \sqrt{(\omega_0^2 - \alpha^2)}$. As $R \rightarrow 0$, $\alpha \rightarrow 0$ and $\omega_d \rightarrow \omega_0$, so the oscillation no longer decays: this idealized, lossless limit is the conceptual seed of every practical frequency-generating circuit discussed later in this report.

2.3 The Source-Free Parallel RLC Circuit

A dual analysis applies to the parallel RLC configuration (Alexander & Sadiku, Section 8.4, p. 324), where Kirchhoff’s current law instead produces a second-order equation in the node voltage, with the same characteristic-equation structure and the same three damping regimes, but with $\alpha = 1 / (2RC)$ for the parallel case. The duality between the series and parallel circuits — voltage and current, R and $1/R$, L and C interchanged — is treated explicitly in the textbook (p. 348) and is useful for recognizing the same oscillatory behavior in different physical arrangements.

2.4 Step Response and the Persistence of Oscillatory Behavior

When a source is suddenly applied to a second-order circuit instead of being removed, the circuit exhibits a step response (Alexander & Sadiku, Sections 8.5 and 8.6, pp. 329 and 334) that still contains the same natural-response term superimposed on a new forced (steady-state) term. In an underdamped circuit, this means the voltage or current rings — oscillates briefly around its final value before settling — a phenomenon visible whenever a switch is closed in a circuit containing both inductance and capacitance, such as an ignition system or a switching power supply. This transient ringing is the same underlying mechanism that, when deliberately reinforced rather than allowed to decay, becomes a continuous oscillator.

3. Frequency Generation

3.1 Resonance as the Frequency-Fixing Mechanism

Resonance is the condition in an RLC network at which the inductive and capacitive reactances cancel, leaving a purely resistive impedance and a current or voltage response that peaks sharply at one frequency. This is treated in Alexander and Sadiku's Chapter 14, "Frequency Response" (Series Resonance and Parallel Resonance sections), and in Boylestad's Introductory Circuit Analysis, Chapter 20, "Resonance" (p. 879). For the series RLC circuit, resonance occurs at

$$\omega_0 = 1 / \sqrt{LC}$$

which is the same natural frequency that appeared in the source-free analysis of Section 2 — confirming that the frequency a circuit oscillates at on its own is exactly the frequency at which it resonates when driven. Two further parameters describe how sharply the response is concentrated around ω_0 : the quality factor Q (the ratio of resonant frequency to bandwidth) and the bandwidth $B = \omega_0/Q$, the width of the frequency band over which the response stays within $\sqrt{2}$ of its peak value (half-power points). A high- Q circuit has a narrow, sharply peaked response and is therefore better suited to generating or selecting one specific frequency with little interference from neighboring frequencies.

Characteristic	Series Resonant Circuit	Parallel Resonant Circuit
Resonance condition	$X_L = X_C$ (reactances cancel)	Susceptance $B_L = B_C$ (imaginary part of $Y = 0$)
Impedance at ω_0	Minimum ($Z = R$)	Maximum ($Z = R$)
Current/Voltage at ω_0	Current is maximum	Voltage is maximum
Typical use	Acceptor / band-pass selection	Rejector circuits, oscillator tank circuits

Table 1. Comparison of series and parallel resonant behavior (Alexander & Sadiku, Ch. 14; Boylestad, Ch. 20, p. 879).

3.2 The LC Tank Circuit

A parallel combination of an inductor and a capacitor, commonly called a tank circuit, is the most direct hardware realization of the natural oscillation discussed in Section 2. An ideal, loss-free tank circuit, once given an initial charge or current, exchanges energy between L and C forever at $f_0 = 1 / (2\pi\sqrt{LC})$ with no external source. Every real tank circuit has finite Q because of winding resistance and dielectric losses, so the oscillation always decays exponentially unless energy is actively replaced — which is exactly the role of the active oscillator circuits discussed next.

3.3 Electronic Oscillators and the Barkhausen Criterion

To convert a damped natural oscillation into a continuous, usable frequency source, an amplifying (active) element must feed energy back into the resonant network in phase with its own oscillation, replacing exactly what is lost to resistance each cycle. The standard condition for sustained oscillation in a feedback amplifier, known as the Barkhausen criterion, requires that the loop gain around the feedback path have a magnitude of exactly one and a phase shift of zero (or an integer multiple of 360°) at the desired frequency of oscillation. Sadiku and Alexander illustrate this principle concretely with a

worked oscillator example in the Applications section of Chapter 10, “Sinusoidal Steady-State Analysis” (Section 10.9.2, “Oscillators”), where an operational-amplifier RC network is shown generating a sustained sinusoidal output from a DC supply by satisfying this feedback condition. Common oscillator families that apply the same underlying principle include:

- RC phase-shift oscillators, which use a chain of resistor–capacitor sections to produce the 180° phase shift needed (combined with an inverting amplifier) to satisfy the Barkhausen condition, typically used for audio-range frequencies.
- The Wien-bridge oscillator, an RC-bridge network feeding a non-inverting amplifier, widely used in low-distortion, audio-frequency function generators.
- LC oscillators such as the Colpitts and Hartley configurations, which use a tank circuit (as in Section 3.2) together with a transistor or op-amp to sustain oscillation at radio frequencies, with the tapped capacitor (Colpitts) or tapped inductor (Hartley) providing the feedback path.
- Crystal oscillators, which replace the LC tank with a piezoelectric quartz crystal whose extremely high Q gives far greater frequency stability, and which form the timing reference for clocks, microprocessors, and communication equipment.

In every case the design task is the same one introduced in Section 2 and 3.1: choose a resonant network to fix the desired frequency ω_0 , and choose an active feedback path that exactly compensates for the network’s losses at that frequency.

3.4 Frequency Response, Transfer Functions, and Bode Plots

Designing or analyzing any frequency-generating or frequency-selecting circuit requires the transfer-function and frequency-response tools developed in Alexander and Sadiku, Chapter 14. The transfer function $H(\omega)$, defined as the frequency-dependent ratio of a phasor output to a phasor input, together with the decibel scale and Bode plots, allows an engineer to predict exactly how the gain and phase of a candidate oscillator or filter network vary with frequency, and to verify that the Barkhausen phase condition is met at the intended operating frequency before a physical circuit is built.

4. AC Signal Generation

4.1 Characterizing the Sinusoidal Signal

Once a frequency has been fixed by a resonant or oscillator network, the resulting signal is described, exactly like any other sinusoid, by three parameters: amplitude, angular frequency, and phase. Alexander and Sadiku formalize the general sinusoid (Chapter 9, “Sinusoids and Phasors,” Section 9.2, p. 369) as

$$v(t) = V_m \cos(\omega t + \varphi)$$

where V_m is the amplitude, $\omega = 2\pi f$ is the angular frequency in radians per second, and φ is the phase angle. Boylestad covers the same waveform parameters — period, frequency, amplitude, and phase relationships between voltage and current — in Introductory Circuit Analysis, Chapter 13, “Sinusoidal Alternating Waveforms” (p. 545), which is the standard starting point for AC circuit theory in that text.

4.2 Phasor Representation

Because every linear circuit driven at a single frequency produces sinusoidal voltages and currents of that same frequency, differing only in amplitude and phase, sinusoids are converted into phasors — complex numbers carrying only magnitude and phase information — to avoid solving differential equations directly. Alexander and Sadiku introduce phasors in Section 9.3 (p. 374) and the corresponding impedance concept in Sections 9.4–9.5 (pp. 383–387), while Boylestad develops the same complex-number (phasor) treatment of resistors, inductors, and capacitors in Chapter 14, “The Basic Elements and Phasors” (p. 595). This phasor/impedance framework is what makes practical AC signal-generation and signal-processing circuits — oscillators, filters, phase shifters, AC bridges — analyzable with ordinary algebra instead of calculus.

4.3 Practical AC Signal Sources

In the laboratory and in industry, AC signals are produced by purpose-built signal or function generators, which are essentially the oscillator circuits of Section 3.3 packaged with adjustable frequency, amplitude, and waveform controls (sine, square, or triangular output). A voltage-controlled oscillator (VCO) extends this idea by allowing the output frequency itself to be varied electronically through a control voltage, which is the basis of frequency modulation (FM) and of phase-locked loops used for frequency synthesis. Related signal-shaping circuits explicitly covered in Alexander and Sadiku’s Applications sections include phase shifters, used to introduce a controlled lag or lead between two AC signals (Section 9.8.1, p. 394), and AC bridges, used to measure or compare unknown impedances against a known AC reference signal (Section 9.8.2, p. 394).

4.4 From Generation to Application

The frequency response and resonance tools of Chapter 14 are also what allow a generated AC signal to be put to use selectively: a radio receiver’s tuning circuit is a parallel resonant network set to the desired carrier frequency, a touch-tone telephone separates different sinusoidal tones using band-pass filters, and a loudspeaker crossover network divides an audio signal into low- and high-frequency bands using the same first- and second-order filter structures developed earlier in the same chapter. In each case the chain is identical: a resonant network establishes a frequency, an active or passive network shapes or filters it, and the resulting AC signal performs a specific communication or control function.

5. Synthesis: How the Three Topics Connect

The three topics of this report are not independent: they form a single causal chain. Oscillation (Section 2) is the natural tendency of any circuit containing both an inductor and a capacitor to exchange energy periodically; this tendency is described mathematically by the same characteristic equation, whether the circuit is left source-free or driven by a fixed source. Frequency generation (Section 3) takes that natural tendency and (a) fixes its rate precisely through resonance, and (b) sustains it indefinitely by adding an active feedback path that exactly cancels resistive losses, in accordance with the Barkhausen criterion. AC signal generation (Section 4) is the practical outcome: a stable, usable sinusoidal (or other periodic) voltage or current, describable by phasors, that can be shaped, filtered, transmitted, or measured using the same frequency-domain tools used to design the oscillator in the first place. Recognizing this single

thread, rather than memorizing the three topics separately, is the central conceptual contribution this report aims to make.

6. Applications and Engineering Relevance

The principles surveyed above underlie a wide range of practical systems:

- Communication systems: carrier-frequency generation, frequency modulation, channel tuning, and intermediate-frequency filtering in radio and television receivers.
- Digital systems and computing: crystal-oscillator clock generation that synchronizes every operation inside a microprocessor or microcontroller.
- Power electronics: switching frequency generation in inverters and switch-mode power supplies, where the oscillation frequency directly determines efficiency and component size.
- Instrumentation: laboratory function generators, AC bridges for impedance measurement, and signal generators used for circuit testing.
- Industrial and biomedical systems: induction heating and ultrasonic transducer drive circuits, both of which depend on tuned resonant networks operating at a precisely generated frequency.

7. Conclusion

Oscillation, frequency generation, and AC signal generation are best understood not as three separate chapters of a circuits course but as three stages of one continuous physical and engineering idea: energy storage elements naturally exchange energy at a characteristic rate; resonance and feedback fix and sustain that rate; and the resulting periodic signal, represented through phasors and analyzed through frequency-response techniques, becomes the working AC signal that powers modern communication, computing, and instrumentation. The second-order circuit analysis, resonance theory, and phasor methods developed in Alexander and Sadiku's *Fundamentals of Electric Circuits* and in Boylestad's *Introductory Circuit Analysis* provide the complete mathematical toolkit needed to design, analyze, and apply every stage of this chain.

References

Alexander, C. K., & Sadiku, M. N. O. Fundamentals of Electric Circuits (6th/7th ed.). McGraw-Hill Education.

Chapter 8, “Second-Order Circuits” — pp. 311–365 (Source-Free Series RLC Circuit, p. 317; Source-Free Parallel RLC Circuit, p. 324; Step Response of a Series RLC Circuit, p. 329; Step Response of a Parallel RLC Circuit, p. 334; Duality, p. 348).

Chapter 9, “Sinusoids and Phasors” — pp. 367–409 (Sinusoids, p. 369; Phasors, p. 374; Phasor Relationships for Circuit Elements, p. 383; Impedance and Admittance, p. 385; Applications — Phase Shifters and AC Bridges, p. 394).

Chapter 10, “Sinusoidal Steady-State Analysis” — beginning p. 411/412 (Applications, Section 10.9, including 10.9.2 “Oscillators”).

Chapter 14, “Frequency Response” (Transfer Function; Decibel Scale; Bode Plots; Series Resonance; Parallel Resonance; Passive and Active Filters; Applications).

Boylestad, R. L. Introductory Circuit Analysis (12th ed.). Pearson/Prentice Hall.

Chapter 13, “Sinusoidal Alternating Waveforms” — p. 545.

Chapter 14, “The Basic Elements and Phasors” — p. 595.

Chapter 15, “Series and Parallel ac Circuits” — p. 645.

Chapter 20, “Resonance” — p. 879.

Note: the page numbers above correspond to the commonly available print editions referenced during preparation of this report (Alexander & Sadiku, 6th/7th ed.; Boylestad, 12th ed.). Pagination can shift by a few pages between printings or regional/international editions, so it is good academic practice to confirm each page number against the specific copy you are citing before final submission.